THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student states: "The HCF of two numbers can be greater than their LCM." The correct response is: 1

(a) Correct **(b)** Correct for primes

(c) Incorrect — HCF ≤ LCM always **(d)** Cannot be determined

2. A student states: "$\frac{1}{\sqrt{2}}$ is rational because it is written as a fraction." The correct response is: 1

(a) Correct **(b)** Correct for even numbers

(c) Incorrect — $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ is irrational **(d)** Cannot be determined

3. If α, β are the zeroes of $x^2 - 2x - 8$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 10 **(b)** −16 **(c)** 16 **(d)** −6

4. If 2 is a zero of $x^2 + ax - 8$, then a is: 1

(a) −4 **(b)** 2 **(c)** −2 **(d)** 4

5. $ax + 2y = 9$ and $3x + by = 18$ have infinitely many solutions when (a, b) is: 1

(a) $(6, 1)$ **(b)** $(3, 4)$ **(c)** $(\frac{3}{2}, 2)$ **(d)** $(\frac{3}{2}, 4)$

6. The pair $x + y = 2$ and $2x + 2y = 4$ has: 1

(a) no solution **(b)** a unique solution

(c) infinitely many solutions **(d)** two solutions

7. The roots of $4x^2 - 12x + 9 = 0$ are: 1

(a) real and equal **(b)** zero

(c) real and distinct **(d)** not real

8. If $x^2 - 6x + k = 0$ has equal roots, then k is: 1

(a) 9 **(b)** 6 **(c)** 36 **(d)** 3

9. A student states: "For the AP $3, 3, 3, \ldots$ the common difference is undefined." The correct response is: 1

(a) Correct **(b)** $d = 3$

(c) Incorrect — $d = 0$ **(d)** $d = 1$

10. If $a_n = 5 - 2n$, Student A says $d = -2$; Student B says $d = 5$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

11. In $\triangle ABC$, $DE \parallel BC$ with $AD = 1$ cm, $DB = 2$ cm and $BC = 6$ cm. Then DE is: 1

(a) 2 cm **(b)** 4 cm **(c)** 3 cm **(d)** 1.5 cm

12. All circles are: 1

(a) equal in area **(b)** congruent

(c) neither similar nor congruent **(d)** similar

13. A student states: "$\sin 60^\circ = 2\sin 30^\circ$." The correct response is: 1

(a) Correct **(b)** Incorrect — $\frac{\sqrt{3}}{2} \neq 1$

(c) Correct for all angles **(d)** Cannot be determined

14. The maximum number of common points of a tangent and a circle is: 1

(a) 0 **(b)** infinitely many **(c)** 1 **(d)** 2

15. The area of a sector of radius 7 cm and angle 60° is: 1

(a) 22 cm² **(b)** $\frac{77}{3}$ cm² **(c)** 77 cm² **(d)** $\frac{154}{3}$ cm²

16. The angle described by the minute hand of a clock in 5 minutes is: 1

(a) 30° **(b)** 6° **(c)** 5° **(d)** 60°

17. A student states: "The surface area of a toy made of a cone on a hemisphere is $\pi rl + 3\pi r^2$." The correct response is: 1

(a) Correct **(b)** Incorrect — it is $\pi rl + 2\pi r^2$

(c) It is $\pi rl + \pi r^2$ **(d)** Cannot be determined

18. The mode of 2, 3, 3, 5, 3, 7 is: 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 7

19. The mean of 5 observations is 18. If one observation 22 is removed, the new mean is: 1

(a) 15 **(b)** 16 **(c)** 18 **(d)** 17

20. A student states: "Experimental and theoretical probabilities are always equal." The correct response is: 1

(a) Correct

(b) Incorrect — they come closer as trials increase

(c) Equal for coins

(d) Cannot be determined

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the HCF of 867 and 255 using prime factorisation. 2

22. Find the zeroes of $x^2 - (\sqrt{3}+1)x + \sqrt{3}$. 2

23. Solve for x: $x^2 - 2ax + a^2 - b^2 = 0$. 2

24. $\triangle ABC \sim \triangle PQR$ with the ratio of corresponding sides 2 : 3. If $PQ = 9$ cm and $QR = 12$ cm, find AB and BC. 2

25. The mid-point of the segment joining A(3, 4) and B(k, 6) is (4, 5). Find k. 2

26. Evaluate: $\frac{\sin^2 45^\circ + \cos^2 45^\circ}{\tan^2 60^\circ}$. 2

27. The angle of elevation of the top of a tower from a point 30 m away is 60°. Find its height. 2

28. TP and TQ are tangents to a circle with centre O and $\angle POQ = 110^\circ$. Find $\angle TPQ$. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $x + y = 5$ and $x - y = 1$. **3**

30. Find the mean by the step-deviation method: **3**

Class	10–30	30–50	50–70	70–90	90–110
Frequency	8	12	20	6	4

31. Find the sum of all multiples of 9 lying between 300 and 700. **3**

32. Prove that the centre of a circle touching two intersecting lines lies on the bisector of the angle formed by the lines. **3**

33. Find the area of the minor segment of a circle of radius 6 cm when the sector angle is $60°$. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) **3**

34. Two dice are thrown. Find P(difference 0), P(sum 6) and P(at least one 4). **3**

35. A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank is $60°$. When he moves 40 m away from the bank, the angle becomes $30°$. Find the height of the tree and the width of the river. **3**

OR

From a point P on the ground, the angle of elevation of the top of a 10 m tall building is $30°$. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is $45°$. Find the length of the flagstaff and the distance of the building from P. **3**

36. Find the median of: **3**

Class	20–25	25–30	30–35	35–40	40–45
Frequency	4	10	16	12	8

OR

A student says the mode of the following data is 6. Verify. **3**

Family size	1–3	3–5	5–7	7–9	9–11
Families	6	9	15	9	1

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Find the ratio in which the line segment joining A(1, −5) and B(−4, 5) is divided by the x-axis. Also find the point of division, and verify using the section formula for the x-coordinate. **5**

OR

Find the coordinates of the point which divides the segment joining A(2, −3) and B(12, 7) in the ratio 2 : 3. If the x-coordinate of A is increased by 4 and the y-coordinate of B is decreased by 5, find the coordinates of the point dividing the new segment in the same ratio. **5**

38. Prove that $(\csc A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$. **5**

OR

Prove that $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\csc\theta$. **5**

39. A solid is composed of a cylinder with hemispherical ends. The whole length of the solid is 108 cm and the diameter of the ends is 36 cm. Find the cost of polishing its surface at 7 paise per cm². **5**

OR

Case study: A circus tent is cylindrical to a height of 3 m and conical above it. The base radius is 52.5 m and the slant height of the cone is 53 m. 5

(i) Find the area of canvas needed.

(ii) Find its cost at ₹12 per m².

40. For the data below, find 5

(i) the mean

(ii) the median

(iii) the mode using the empirical relationship.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	4	8	10	12	6

OR

Case study: A student says the mode of the data below is 42. 5

(i) Find the actual mode.

(ii) Find the median.

(iii) Is the student correct?

Class	0–20	20–40	40–60	60–80	80–100	100–120	120–140
Frequency	6	8	10	12	6	5	3

END OF QUESTION PAPER **Total: 80 Marks**

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